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Chapter 16

Managing Bond Portfolios

INVESTMENTS

THIRTEENTH EDITION

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Chapter Overview

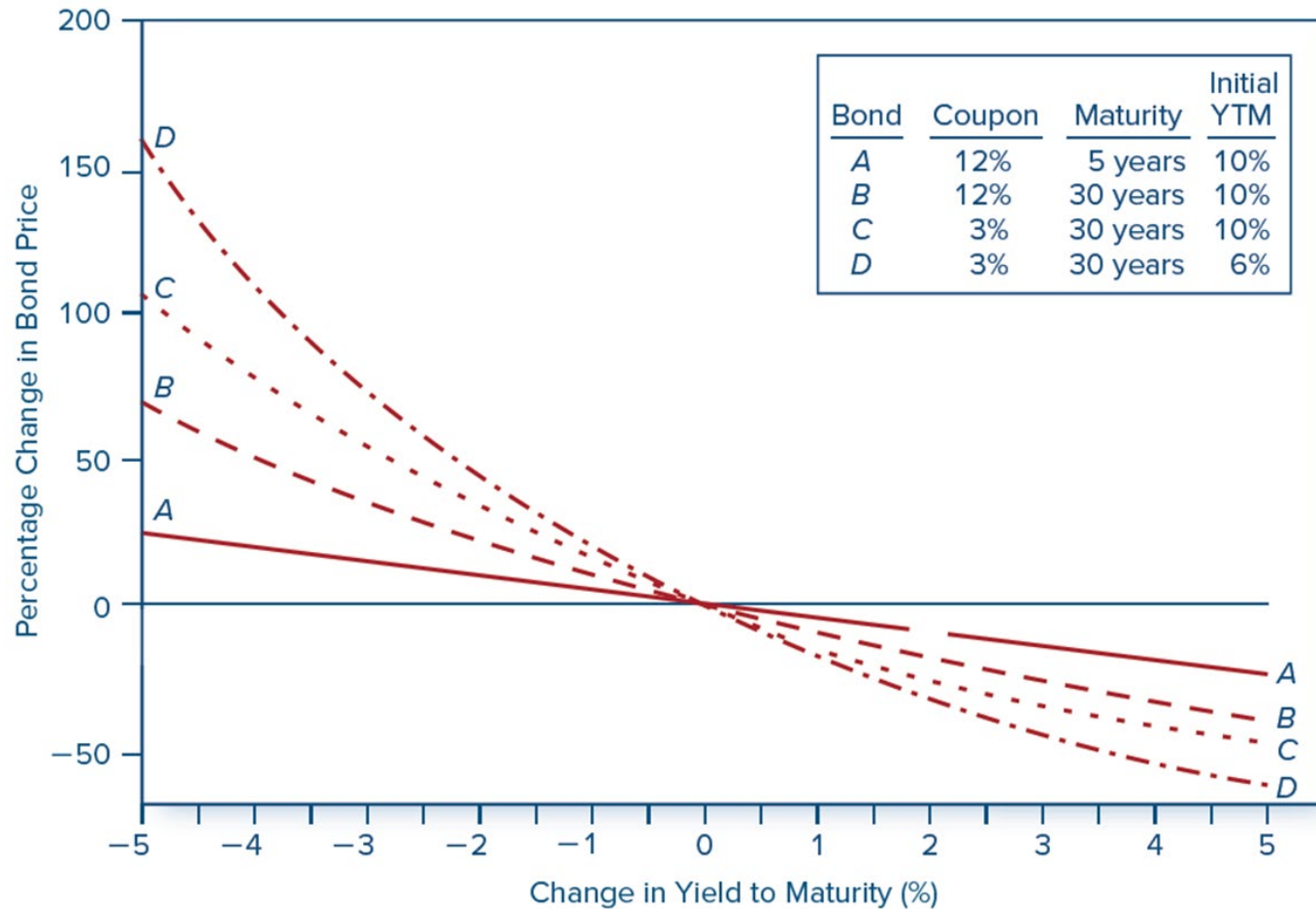
- Examine various fixed-income portfolio strategies.
 - Distinguish between passive and active approaches.
- Discuss sensitivity of bond prices to interest rates fluctuations.
 - Sensitivity is measured by duration.
- Consider refinements in the way interest rate sensitivity is measured, focusing on bond convexity.

Malkiel's Bond–Pricing Relationships

1

1. Bond prices and yields are inversely related.
2. An increase in a bond's yield to maturity results in a smaller price change than a decrease in yield of equal magnitude.
3. Prices of long-term bonds tend to be more sensitive to interest rate changes than prices of short-term bonds.

Figure 16.1 Change in Bond Price as a Function of Change in Yield to Maturity



Malkiel's Bond–Pricing Relationships

2

- 4. Interest rate risk is less than proportional to bond maturity.
- 5. Interest rate risk is inversely related to the bond's coupon rate.
- 6. The sensitivity of a bond's price to a change in its yield is inversely related to the YTM at which the bond is currently selling (Homer and Liebowitz).

Table 16.1 Prices of 8% Coupon Bond (Coupons Paid Semi-Annually)

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Table 16.1

Prices of 8% coupon bond
(coupons paid semiannually)

Yield to Maturity (APR)	<i>T</i> = 1 Year	<i>T</i> = 10 Years	<i>T</i> = 20 Years
8%	1,000.00	1,000.00	1,000.00
9%	990.64	934.96	907.99
Fall in price (%)*	0.94%	6.50%	9.20%

*Equals value of bond at a 9% yield to maturity divided by value of bond at (the original) 8% yield, minus 1.

*Equals value of bond at a 9% yield to maturity divided by value of bond at (the original) 8% yield, minus 1.

Table 16.2 Prices of Zero-Coupon Bond (Semi-Annual Compounding)

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Table 16.2

Prices of zero-coupon bond
(semiannual compounding)

Yield to Maturity (APR)	<i>T</i> = 1 Year	<i>T</i> = 10 Years	<i>T</i> = 20 Years
8%	924.56	456.39	208.29
9%	915.73	414.64	171.93
Fall in price (%)*	0.96%	9.15%	17.46%

*Equals value of bond at a 9% yield to maturity divided by value of bond at (the original) 8% yield, minus 1.

*Equals value of bond at a 9% yield to maturity divided by value of bond at (the original) 8% yield, minus 1.

Duration

- A measure of the average maturity of a bond's promised cash flows.
- **Macaulay's duration** equals the weighted average of the times to each coupon or principal payment.
 - Weight applied to each payment time is proportion of total value of bond accounted for by that payment (i.e., the PV of the payment divided by the bond price).
- Duration = Maturity for zero coupon bonds.
- Duration < Maturity for coupon bonds.

Duration Calculation

- Duration calculation:

$$D = \sum_{t=1}^T t \times w_t$$

$$w_t = \frac{CF_t / (1 + y)^t}{P}$$

CF_t = Cash Flow at Time t

P = Price of Bond

y = Yield to Maturity

Spreadsheet 16.1 Calculating the Duration of Two Bonds

	A	B	C	D	E	F	G
1			Time until		PV of CF		Column (C)
2			Payment		(Discount rate =		times
3		Period	(Years)	Cash Flow	5% per period)	Weight*	Column (F)
4	A. 8% coupon bond	1	0.5	40	38.095	0.0395	0.0197
5		2	1.0	40	36.281	0.0376	0.0376
6		3	1.5	40	34.554	0.0358	0.0537
7		4	2.0	1040	855.611	0.8871	1.7741
8	Sum:				964.540	1.0000	1.8852
9							
10	B. Zero-coupon	1	0.5	0	0.000	0.0000	0.0000
11		2	1.0	0	0.000	0.0000	0.0000
12		3	1.5	0	0.000	0.0000	0.0000
13		4	2.0	1000	822.702	1.0000	2.0000
14	Sum:				822.702	1.0000	2.0000
15							
16	Semiannual int rate:	0.05					
17							
18	*Weight = Present value of each payment (column E) divided by the bond price.						

Spreadsheet 16.1 Calculating the duration of two bonds Column sums subject to rounding error.

Interest Rate Risk ₁

- Duration as a measure of interest rate sensitivity.
 - Price change is proportional to duration.

$$\frac{\Delta P}{P} = -D \times \left[\frac{\Delta(1+y)}{1+y} \right]$$

- $D^* = D/(1+y)$ = **Modified duration.**

$$\frac{\Delta P}{P} = -D^* \Delta y$$

Interest Rate Risk ₂

- Consider the 2-year maturity, 8% coupon bond in Spreadsheet 16.1 making semiannual coupon payments and selling at \$964.54, with a yield to maturity of 10% and duration of 1.8852.
- What is the approximate price change if the yield to maturity increases to 10.5%?

$$\frac{\Delta P}{P} = -D^* \Delta y \rightarrow$$

$$\Delta P = -D^* \Delta y \times P$$

$$= -\frac{1.8852}{1.10} \times .005 \times \$964.54 = \$8.27$$

Duration and Interest Rate Risk ₁

- Two bonds have duration of 1.8852 years
 - Bond A: 2-year, 8% coupon bond with YTM = 10%
 - Bond B: Zero coupon bond maturing in 1.8852 years
- Duration of both bonds is $1.8852 \times 2 = 3.7704$ semiannual periods. $y = 5\%$
- Modified $D = 3.7704 / (1 + 0.05) = 3.591$ periods

Duration and Interest Rate Risk ₂

- Suppose the semiannual interest rate increases by 0.01%. Bond prices fall by

$$\frac{\Delta P}{P} = -D * \Delta y$$

$$= -3.591 \times 0.01\%$$

$$= -0.03591\%$$

- Bonds with equal D → same interest rate sensitivity

Duration and Interest Rate Risk ³

Coupon Bond

The coupon bond price drops from \$964.540 to \$964.1942, when its yield increases to 5.01%

Percentage decline of 0.0359%

Zero

The zero-coupon bond price drops from \$831.9704 ($\$1,000/1.05^{3.7704}$) to \$831.6717 ($\$1,000/1.05^{01}_{01}{}^{3.7704}$)

Percentage decline of 0.0359%

Duration Rules ¹

Rule 1

- The duration of a zero-coupon bond equals its time to maturity.

Rule 2

- Holding maturity constant, a bond's duration is lower when the coupon rate is higher.

Rule 3

- Holding the coupon rate constant, a bond's duration generally increases with its time to maturity.

Cash Flows Paid by 9% Coupon, Annual Payment Bond with an 8-Year Maturity and 10% Yield to Maturity

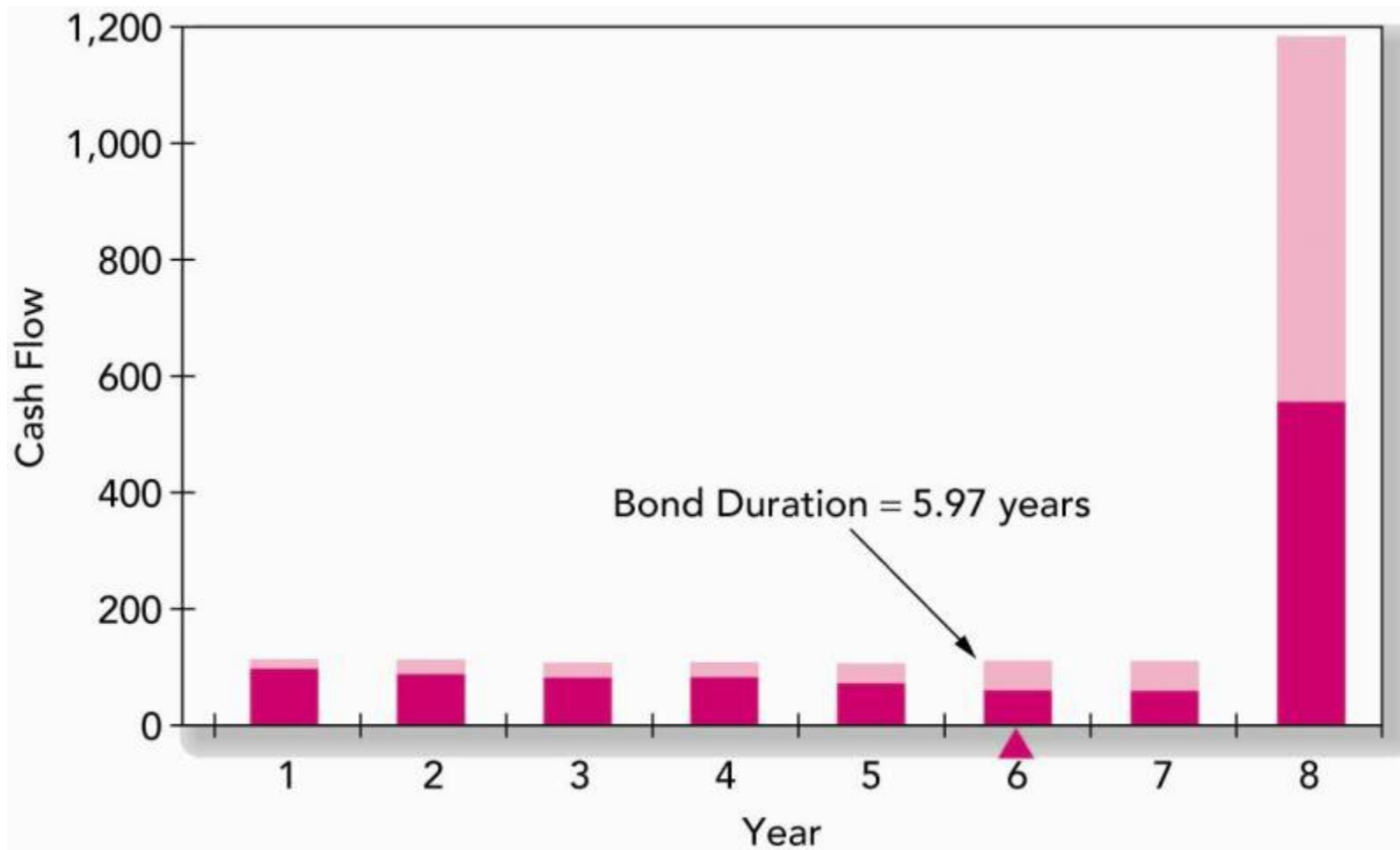
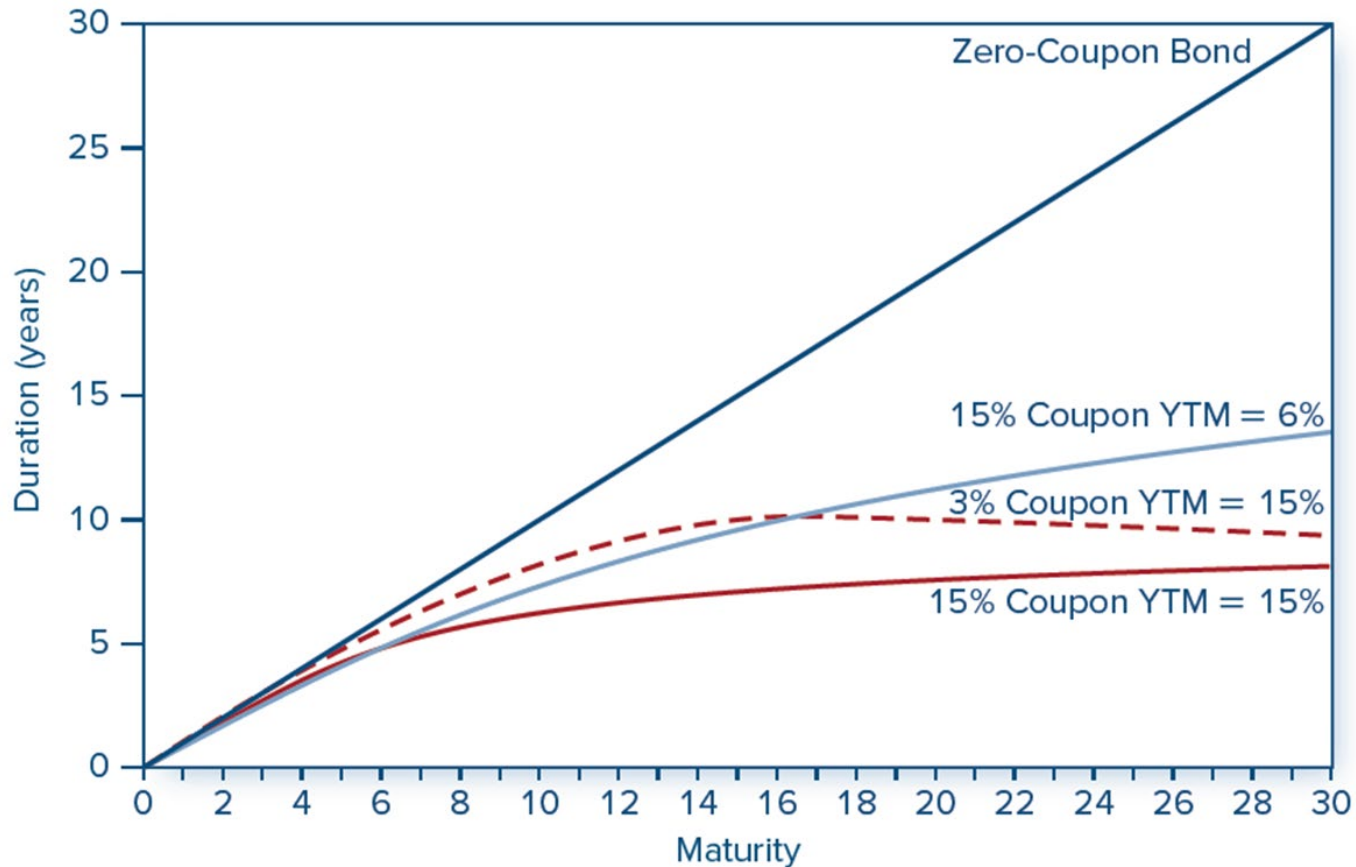


Figure 16.2 Bond Duration Versus Bond Maturity



Duration Rules ²

Rule 4

- Holding other factors constant, the duration of a coupon bond is higher when the bond's yield to maturity is lower

Rule 5

- The duration of a level perpetuity is equal to:

$$\frac{1 + y}{y}$$

[Handout](#) for derivation

Table 16.3 Bond Durations

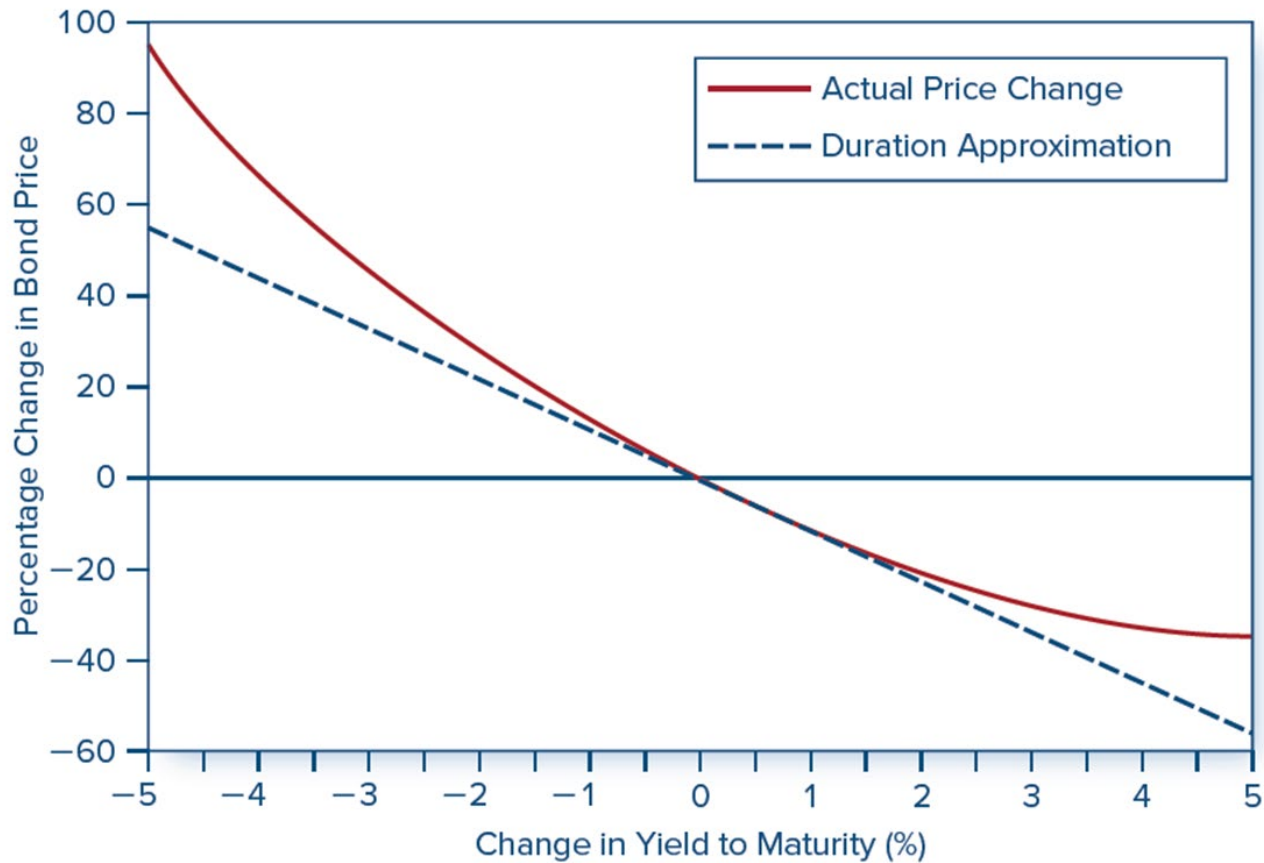
(Yield to Maturity = 8% APR; Semi-Annual Coupons)

Years to Maturity	Coupon Rates (per Year)			
	6%	8%	10%	12%
1	0.985	0.981	0.976	0.972
5	4.361	4.218	4.095	3.990
10	7.454	7.067	6.772	6.541
20	10.922	10.292	9.870	9.568
Infinite (perpetuity)	13.000	13.000	13.000	13.000

Convexity ¹

- Relationship between bond prices and yields is not linear.
- Duration rule is a good approximation for only small changes in bond yields.
- Bonds with higher **convexity** exhibit higher curvature in the price–yield relationship.
 - Convexity is measured as the rate of change of the slope of the price–yield curve, expressed as a fraction of the bond price.

Figure 16.3 Bond Price Convexity (30-Year Maturity; 8% Coupon; Initial YTM = 8%)



Convexity²

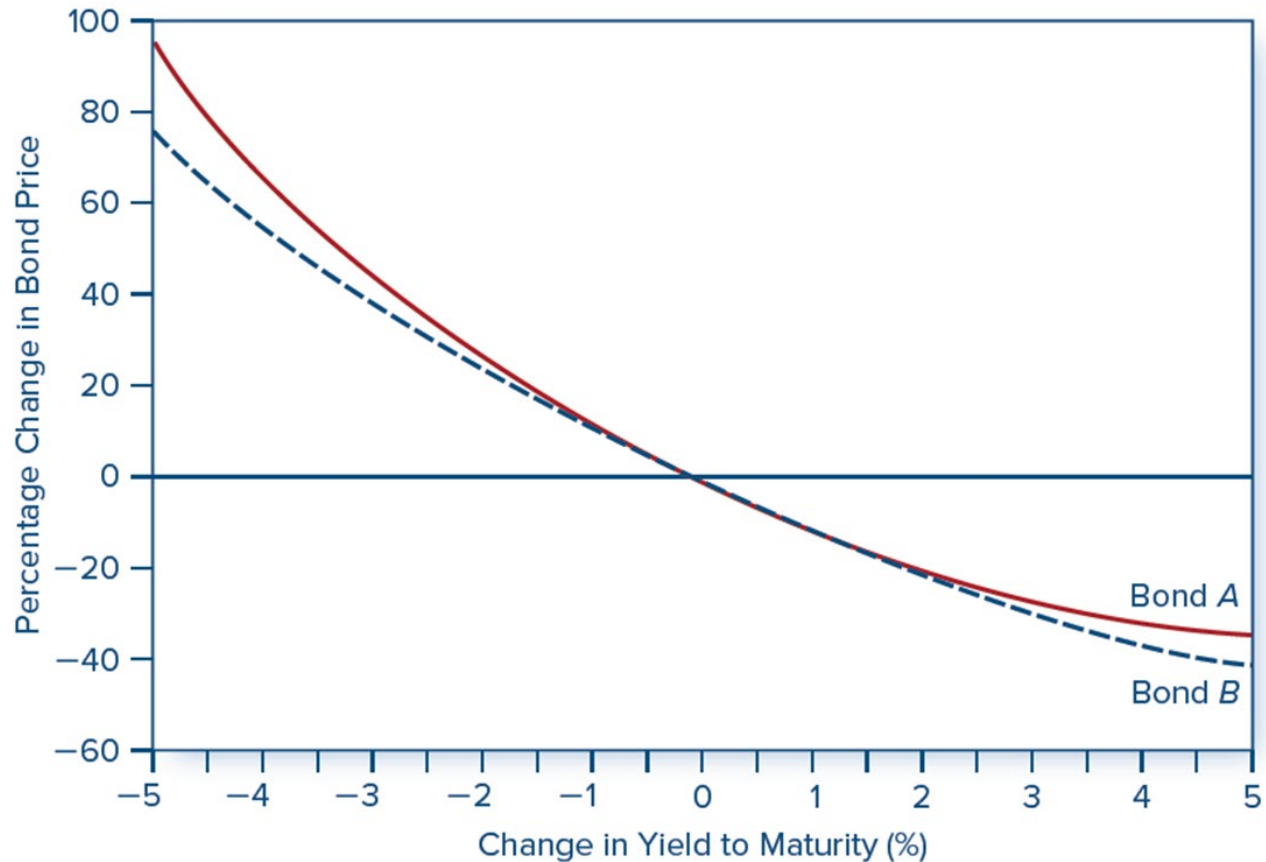
$$\text{Convexity} = \frac{1}{P \times (1 + y)^2} \sum_{t=1}^T \left[\frac{CF_t}{(1 + y)^t} (t^2 + t) \right]$$

- Accounting for convexity changes the equation:

$$\frac{\Delta P}{P} = -D^* \Delta y + \frac{1}{2} \times \text{Convexity} \times (\Delta y)^2$$

[Handout](#) for details

Figure 16.4 Convexity of Two Bonds



Why Do Investors Like Convexity?

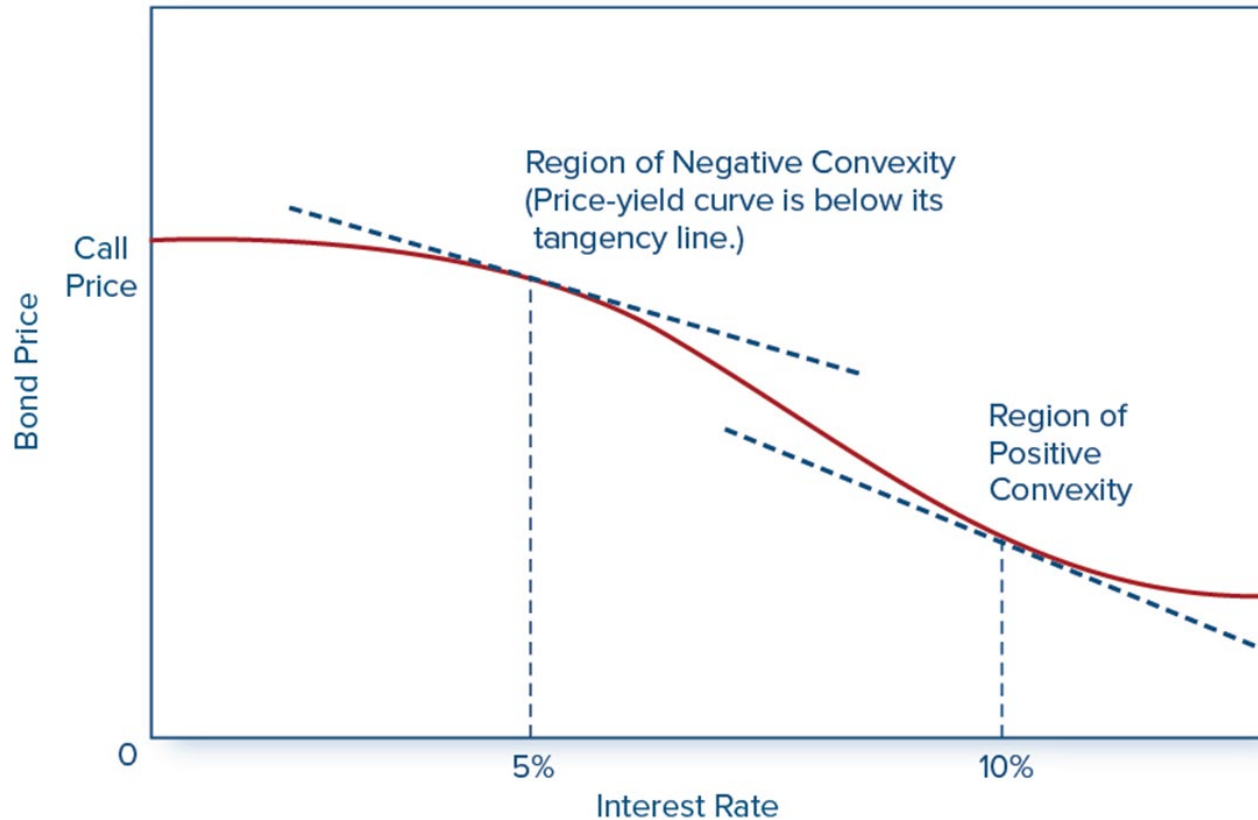
- Bonds with greater curvature gain more in price when yields fall than they lose when yields rise.
 - The more volatile interest rates, the more attractive this asymmetry.
- Investors must pay higher prices and accept lower yields to maturity on bonds with greater convexity.

Duration and Convexity of Callable Bonds

- As rates fall, there is a ceiling on the bond's market price, which cannot rise above the call price.
- As rates fall, the bond is subject to price compression.
- Use effective duration:

$$\text{Effective Duration} = -\frac{\Delta P/P}{\Delta r}$$

Figure 16.5 Price–Yield Curve for a Callable Bond



Example 16.3 Effective Duration

- Suppose that a callable bond with a call price of \$1,050 is selling today for \$980. If the yield curve shifts up by .5%, the bond price will fall to \$930. If it shifts down by .5%, the bond price will rise to \$ 1,010. To compute effective duration, we compute:

$$\Delta r = \text{Assumed increase in rates} - \text{Assumed decrease in rates} = .5\% - (-.5\%) = 1\% = .01$$

$$\Delta P = \text{Price at .5\% increase in rates} - \text{Price at .5\% decrease in rates} = 930 - (1,010) = -\$80$$

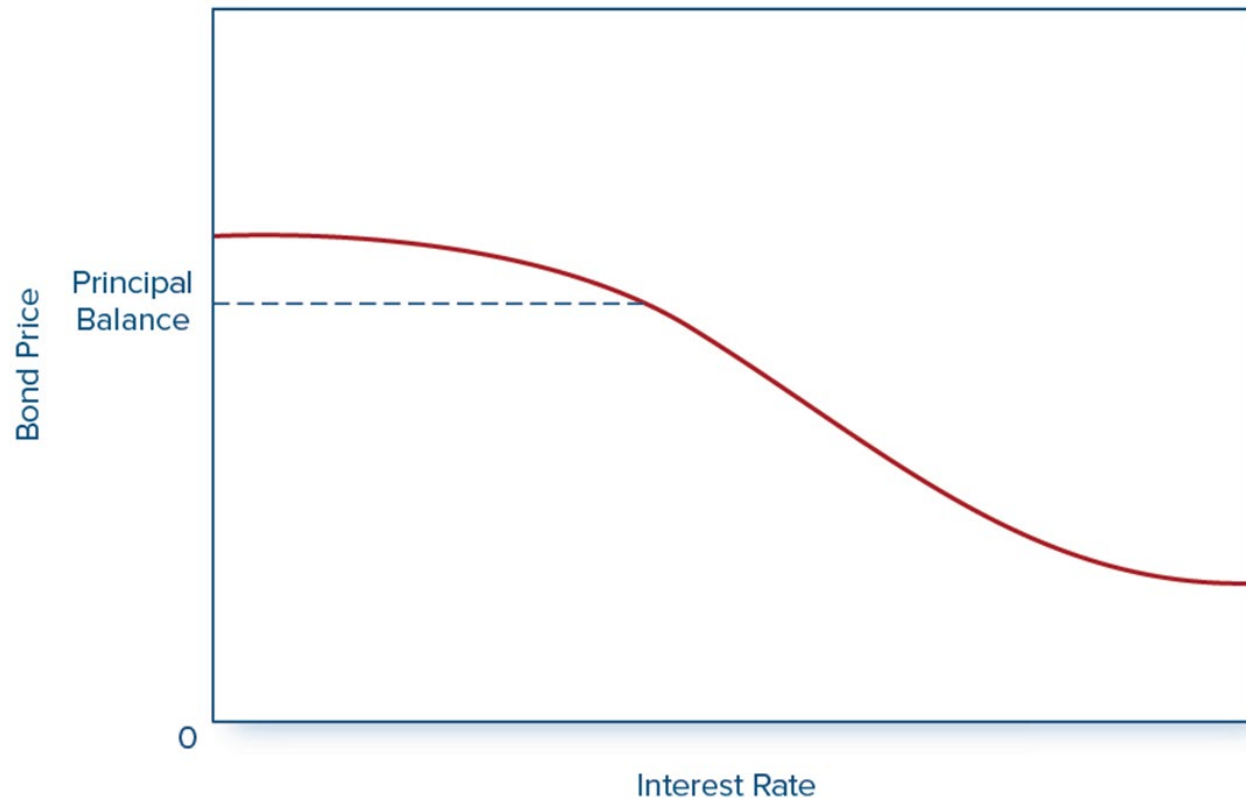
Example 16.3 Effective Duration (cont.)

- Then the effective duration of the bond is:
 - Effective duration =
$$- \{(-\$80) / \$980\} / .01 = 8.16 \text{ years}$$
- In other words, the bond price changes by 8.16% for a 1 percentage point swing in rates around current values.

Duration and Convexity: MBS

- Mortgage-Backed Securities (MBS).
- Though the number of outstanding callable corporate bonds has declined, the MBS market has grown rapidly.
- MBS are a portfolio of callable amortizing loans.
 - Homeowners may repay their loans at any time.
 - MBS have negative convexity.
 - Often sell for more than their principal balance.
 - Homeowners do not refinance as soon as rates drop, so implicit call price is not a firm ceiling on MBS value.

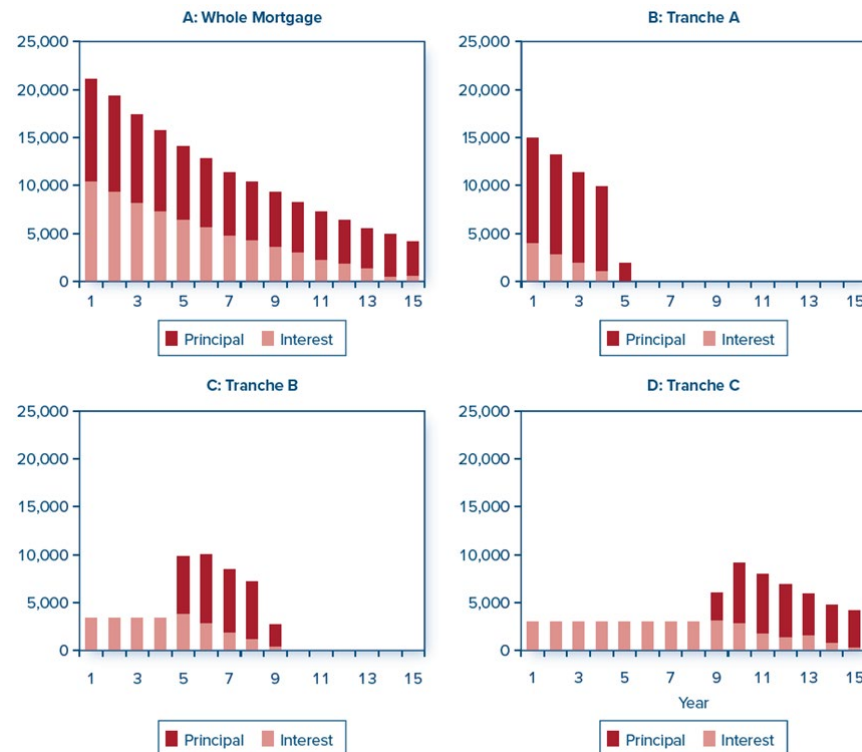
Figure 16.6 Price–Yield Curve for a Mortgage-Backed Security



Duration and Convexity: CMO

- Collateralized Mortgage Obligation (CMO).
 - Further redirects the cash flow stream of the MBS to several classes of derivative securities called “tranches.”
 - Tranches may ***allocate interest rate*** risk to investors most willing to bear that risk.
 - Different tranches may receive different coupon rates.
 - Some may be given preferential treatment in terms of uncertainty over mortgage prepayment speeds.

Figure 16.7 CMO Cash Flows



Tranches — the underlying mortgage pool is divided into a set of derivative securities

- Reallocate interest rate risk

Tranche A = \$4 million principal	“Short-pay” tranche
Tranche B = \$3 million principal	“Intermediate-pay” tranche
Tranche C = \$3 million principal	“Long-pay” tranche

Figure 16.7 Panel A: Cash flows to whole mortgage pool; **Panels B–D:** Cash flows to three tranches.

Interest Rate Risk

- Interest rate risk is the possibility that an investor does not earn the promised YTM because of interest rate changes.
- A bond investor faces two types of interest rate risk:
 - **Price risk:** The risk that an investor cannot sell the bond for as much as anticipated. An increase in interest rates reduces the sale price.
 - **Reinvestment risk:** The risk that the investor will not be able to reinvest the coupons at the promised yield rate. A decrease in interest rates reduces the future value of the reinvested coupons.
- The two types of risk are potentially offsetting.

Passive Bond Management

- Passive managers take bond prices as fairly set and seek to control only the risk of their fixed-income portfolio.
- Two classes of passive management:
 - Bond Indexing strategy.
 - Immunization techniques.
- Both classes accept market prices as being correct but differ greatly in terms of risk exposure.

Bond-Index Funds

- Similar to stock market indexing.
 - Create a portfolio that mirrors the composition of an index that measures the broad market .
 - Challenges in construction:
 - Very difficult to purchase each security in the index in proportion to its market value.
 - Many bonds are very thinly traded.
 - Difficult rebalancing problems.
- Due to challenges, a *cellular* approach is pursued (Table 16.8).

Figure 16.8 Stratification of Bonds Into Cells

Term to Maturity \ Sector							
	Treasury	Agency	Mortgage-Backed	Industrial	Finance	Utility	Yankee
<1 year	12.1%						
1–3 years	5.4%						
3–5 years			4.1%				
5–7 years							
7–10 years		0.1%					
10–15 years							
15–30 years			9.2%			3.4%	
30+ years							

Bonds falling within each cell are considered reasonably homogeneous

Passive Management: Indexing

Bond Index Funds

Contains Thousands of Issues, many of which are infrequently traded



Turnover more than stock indexes as the bonds mature



Establish a bond portfolio with representation for each cell that matches the weight of that cell in the bond universe

They only hold a representative sample of the bonds in the actual index

Composition of bond indexes is constantly changing. (new bond issue; unlike new stock issue)

Passive Management: Immunization

- **Immunization** techniques are used to shield overall financial status from interest rate risk.
 - Widely used by pension funds, insurers, and banks.
- Duration-matched assets and liabilities let the asset portfolio meet the firm's obligations despite interest rate movement.
 - Balances reinvestment rate risk and price risk.
 - **Rebalancing** is required to realign the portfolio's duration with the duration of the obligation.

Target Date Immunization

- An insurance company that issues a guaranteed investment contract, or GIC, for \$10,000.
- If the GIC has a 5-year maturity and a guaranteed interest rate of 8%, the insurance company is obligated to pay $\$10,000(1.08)^5 = \$14,693.28$ in 5 years.
- Suppose the insurance company chooses to fund its obligation with \$10,000 of 8% annual coupon bonds, selling at par value, with 6 years to maturity.

Terminal Value of a 6-year Maturity Bond Portfolio After 5 Years

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Payment Number	Years Remaining until Obligation	Accumulated Value of Invested Payment		
A. Rates Remain at 8%				
1	4	$800 \times (1.08)^4$	=	\$ 1,088.39
2	3	$800 \times (1.08)^3$	=	1,007.77
3	2	$800 \times (1.08)^2$	=	933.12
4	1	$800 \times (1.08)^1$	=	864.00
5	0	$800 \times (1.08)^0$	=	800.00
Sale of bond	0	10,800/1.08	=	<u>10,000.00</u>
				\$14,693.28
B. Rates Fall to 7%				
1	4	$800 \times (1.07)^4$	=	\$ 1,048.64
2	3	$800 \times (1.07)^3$	=	980.03
3	2	$800 \times (1.07)^2$	=	915.92
4	1	$800 \times (1.07)^1$	=	856.00
5	0	$800 \times (1.07)^0$	=	800.00
Sale of bond	0	10,800/1.07	=	<u>\$10,093.46</u>
				\$14,694.05
C. Rates Increase to 9%				
1	4	$800 \times (1.09)^4$	=	\$ 1,129.27
2	3	$800 \times (1.09)^3$	=	1,036.02
3	2	$800 \times (1.09)^2$	=	950.48
4	1	$800 \times (1.09)^1$	=	872.00
5	0	$800 \times (1.09)^0$	=	800.00
Sale of bond	0	10,800/1.09	=	<u>9,908.26</u>
				\$14,696.02

Table 16.4

Terminal value of a 6-year-maturity bond portfolio after five years (all proceeds reinvested)

Note: The sale price of the 6-year-maturity bond portfolio equals the portfolio's final payment (\$10,800) divided by $1 + r$ because the time to maturity of the bonds will be 1 year at the time of sale.

Target Date Immunization (cont.)

- For an investor's horizon equal to the portfolio's duration, price risk and reinvestment risk exactly cancel out.
- In the previous example, the duration of the 6-year maturity bonds used to fund the GIC is 5 years.

Figure 16.9 Growth of Invested Funds

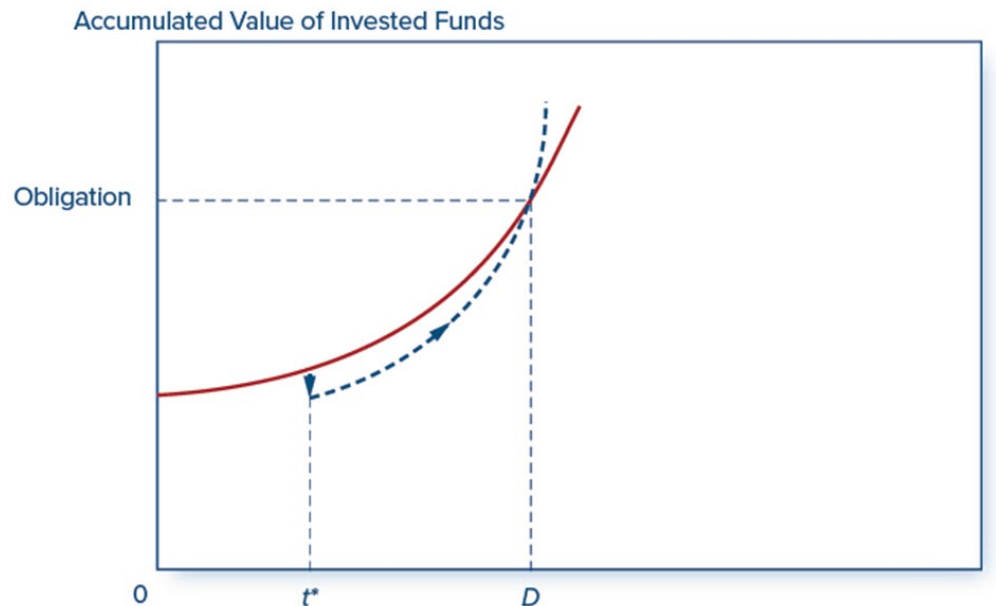


Figure 16.9 Growth of invested funds. The solid-colored curve represents the growth of portfolio value at the original interest rate. If interest rates increase at time V , the portfolio value initially falls but increases thereafter at the faster rate represented by the broken curve. At time D (duration), the curves cross.

Table 16.5 Market Value Balance Sheet

Assets		Liabilities	
A. Interest Rate = 8%			
Bonds	\$10,000	Obligation	\$10,000
B. Interest Rate = 7%			
Bonds	\$10,476.65	Obligation	\$10,476.11
C. Interest Rate = 9%			
Bonds	\$ 9,551.41	Obligation	\$ 9,549.62

Value of bonds = $\$800 \times \text{Annuity factor}(r, 6) + \$10,000 \times \text{PV factor}(r, 6)$

$$\text{Value of obligation} = \frac{14,693.28}{(1+r)^5} = 14,693.28 \times \text{PV factor}(r, 5)$$

Figure 16.10 Immunization

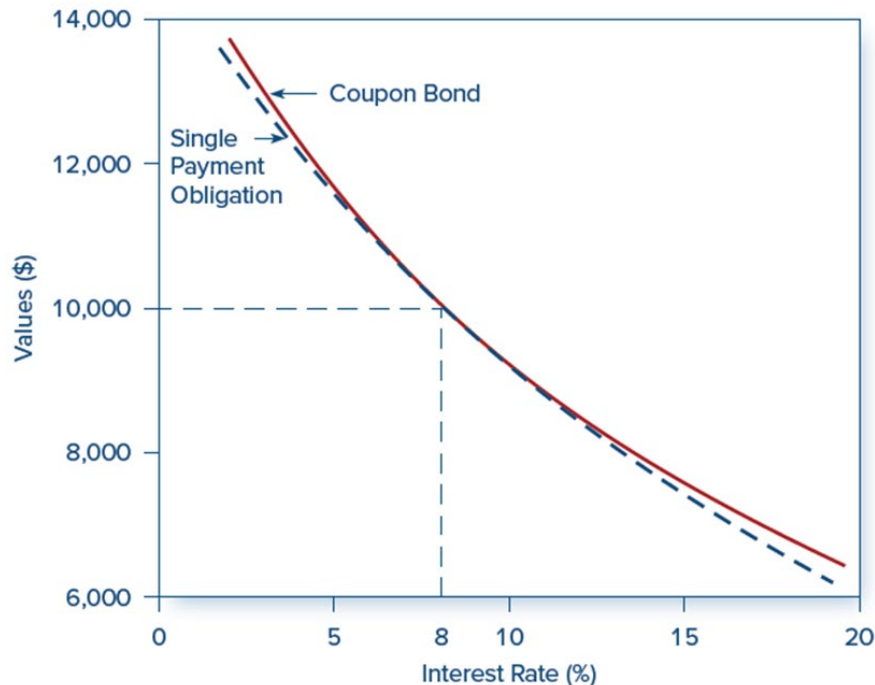


Figure 16.10 Immunization. The coupon bond fully funds the obligation at an interest rate of 8%. Moreover, the present value curves are tangent at 8%, so the obligation will remain fully funded even if rates change by a small amount.

Duration of the Bond Portfolio

- The duration of the portfolio is the weighted average of the durations of the individual bonds in the portfolio:

Portfolio Duration =

$$w_1D_1 + w_2D_2 + w_3D_3 + \dots + w_kD_k$$

where

w_i = market value of bond i / market value of the portfolio

D_i = duration of bond i

k = number of bonds in the portfolio

Constructing an Immunized Portfolio (Example 16.4)

- An insurance company must make a payment of \$19,487 in 7 years. The market interest rate is 10%, so the present value of the obligation is \$10,000.
- The company's portfolio manager wishes to fund the obligation using 3-year zero-coupon bonds and perpetuities paying annual coupons (to keep the algebra simple). How can the manager immunize the obligation?

Constructing an Immunized Portfolio (cont.)

- Immunization requires that the duration of the portfolio of assets equal the duration of the liability.
1. *Calculate the duration of the liability.* In this case, the liability duration is simple to compute. It is a single-payment obligation with duration of 7 years.

Constructing an Immunized Portfolio (cont.)

2. *Calculate the duration of the asset portfolio.* The portfolio duration is the weighted average of duration of each component asset, with weights proportional to the funds placed in each asset.

- The duration of the zero-coupon bond is simply its maturity, 3 years.
- The duration of the perpetuity is $1.10/.10 = 11$ years.
- Therefore, if the fraction of the portfolio invested in the zero is called w , and the fraction invested in the perpetuity is $(1 - w)$, the portfolio duration will be
- Asset duration = $w \times (3 \text{ years}) + (1-w) \times (11 \text{ years})$

Constructing an Immunized Portfolio (cont.)

3. *Find the asset mix that sets the duration of assets equal to the 7-year duration of liabilities.*

- $w \times (3 \text{ years}) + (1-w) \times (11 \text{ years}) = 7 \text{ years}$
- $w = 0.5$

4. *Fully fund the obligation.* Because the obligation has a present value of \$10.000, and the fund will be invested equally in the zero and the perpetuity, the manager must purchase \$5.000 of the zero-coupon bond and \$5,000 of the perpetuity. (Note that the *face value* of the zero will be $\$5.000 \times (1.10)^3 = \$6.655.$)

Rebalancing (Example 16.5)

- Suppose that a year has passed, and the interest rate remains at 10%. The portfolio manager of Example 16.4 needs to reexamine her position.
- Is the position still fully funded? Is it still immunized?
- If not, what actions are required?

Rebalancing (Example 16.5)(cont.)

- First, examine funding.
 - The present value of the obligation will have grown to \$11,000, as it is one year closer to maturity. The manager's funds also have grown to \$11,000:

The zero-coupon bonds have increased in value from \$5,000 to \$5,500 with the passage of time, while the perpetuity has paid its annual \$500 coupons and remains worth \$5,000. Therefore, the obligation is still fully funded.

Rebalancing (Example 16.5) (cont.)

- The portfolio weights must be changed, however.
 - The zero-coupon bond now has a duration of 2 years, while the perpetuity duration remains at 11 years. The obligation is now due in 6 years.
 - The weights must now satisfy the equation

$$w \times 2 + (1 - w) \times 11 = 6$$

which implies that $w = 5/9$. To rebalance the portfolio and maintain the duration match, the manager now must invest a total of $\$11,000 \times 5/9 = \$6,111.11$ in the zero-coupon bond.

This requires that the entire \$500 coupon payment be invested in the zero, with an additional \$111.11 of the perpetuity sold and invested in the zero-coupon bond.

Uses of Duration

- Summary measure of length or effective maturity for a portfolio
- Immunization of interest rate risk (passive management)
 - Net worth immunization
 - Target date immunization
- Measure of price sensitivity for changes in interest rate

Cash Flow Matching and Dedication

- **Cash flow matching** is immunization that requires matching cash flows from a bond portfolio with those of an obligation.
 - Imposes many constraints on bond selection process.
- Cash flow matching in a multiperiod basis is referred to as a **dedication strategy**.
 - Manager selects either zero-coupon or coupon bonds with total cash flows in each period that match a series of obligations.
 - Once-and-for-all approach to eliminating interest rate risk.

Active Bond Management: Sources of Potential Profit

1. **Substitution swap** – exchange of one bond for another more attractively priced bond with similar attributes
2. **Intermarket spread swap** – switch from one segment of the bond market to another (example, Treasuries to corporates)
3. **Rate anticipation swap** – switch made between bonds of different durations in response to forecasts of interest rates
 - Rate increase: shorter duration
 - Rate decrease: longer duration
4. **Pure yield pickup swap** – moving to higher-yield, longer-term bonds to capture the liquidity premium
5. **Tax swap** – swapping two similar bonds to capture a tax benefit

Active Bond Management: Horizon Analysis

- **Horizon analysis** involves forecasting the realized compound yield over various holding periods of investment horizons.
 - Analyst selects a particular holding period and predicts the yield curve at the end of the period.
 - Given a bond's time to maturity at the end of the holding period, its yield can be read from the predicted yield curve and its end-of-period price calculated.

Example 16.6 Horizon Analysis ¹

- A 20-year maturity bond with a 10% coupon rate (paid annually) currently sells at a yield to maturity of 9%. A portfolio manager with a 2-year horizon needs to forecast the total return on the bond over the coming 2 years. In 2 years, the bond will have an 18-year maturity. The analyst forecasts that 2 years from now, 18-year bonds will sell at yields to maturity of 8%, and that coupon payments can be reinvested in short-term securities over the coming 2 years at a rate of 7%.
- To calculate the 2-year return on the bond, the analyst would perform the following calculations:
 1. Current price = \$ 100 × Annuity factor (9%, 20 years)
+ \$1,000 × PV factor (9%, 20 years)
= \$1,091.29
 2. Forecast price = \$100 × Annuity factor (8%, 18 years)
+ \$1,000 × PV factor (8%, 18 years)
= \$1,187.44

Example 16.6 Horizon Analysis ₂

3. The future value of reinvested coupons = $(\$100 \times 1.07) + \$100 = \$207$.

4. The 2-year return = $\frac{\$207 + (\$1,187.44 - \$1,091.29)}{\$1,091.29} = 0.278$, or 27.8%

The annualized rate of return over the 2-year period would then be $1.278^{1/2} - 1 = 0.13$, or 13%.